Report Generation Scripts ReadMe

Scripts

analyse_RH_run.tcl
generate_report.pl

Location

/home/vinayakam/scripts/report_generation/
On ftp : ftp.apache-da.com/outgoing/report_generation_scripts.tar.gz

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Purpose

- Publishing RedHawk analysis results
- Debugging RedHawk run
- Comparison/regression of different RedHawk runs
- And in general, Summarizing a RedHawk run

Flow/Usage

Following diagram shows the RedHawk flow (blue shows the additional operation to be done) and the corresponding commands (green shows the additional command to be executed).

RedHawk Flow	Commands
Load the tcl script	`source analyse_RH_run.tcl`
Perform static analysis	setup design ..to.. perform static
Generate Static reports	`generate_analysis_report -static 1`
Perform dynamic	perform dynamic
Generate Dynamic reports	`generate_analysis_report -dynamic 1`
Perform Rampup analysis	perform lowpower
Generate Rampup reports	`generate_analysis_report -rampup 1`

Modify your RedHawk run tcl files to include the commands highlighted in blue as follows:

run_static.tcl

```
source
/home/vinayakam/scripts/report_generation/analyse_RH_run.tcl
import gsr design.gsr
setup design
perform pwrcalc
perform extraction -power -ground
perform static
generate_analysis_report -static 1
```

run_dynamic.tcl

```
source
/home/vinayakam/scripts/report_generation/analyse_RH_run.tcl
import gsr design.gsr
setup design
perform pwrcalc
perform extraction -power -ground -c
perform dynamic
generate_analysis_report -dynamic 1
```

run_rampup.tcl

```
source
/home/vinayakam/scripts/report_generation/analyse_RH_run.tcl
import gsr design.gsr
setup design
setup analysis_mode lowpower
perform pwrcalc
perform extraction -power -ground -c
perform dynamic -lowpower
generate_analysis_report -rampup 1
```

run_static_dynamic_rampup.tcl (if you are running static,dynamic,rampup in a single run itself)

```
source
/home/vinayakam/scripts/report_generation/analyse_RH_run.tcl
import gsr design.gsr
setup design
perform pwrcalc
perform extraction -power -ground -c
perform static
generate_analysis_report -static 1
perform dynamic
generate_analysis_report -dynamic 1
perform dynamic -lowpower
generate_analysis_report -rampup 1
```

Output

- Analysis Summary Report in HTML and Text formats
  - Viewable in browser
  - Click and view maps and movie in browser itself
  - Click and open report/log files
- Other text reports
  - Gridcheck report
  - Decap statistics report
- Maps
  - Voltage drop maps
  - Current and EM related maps
  - Power Maps

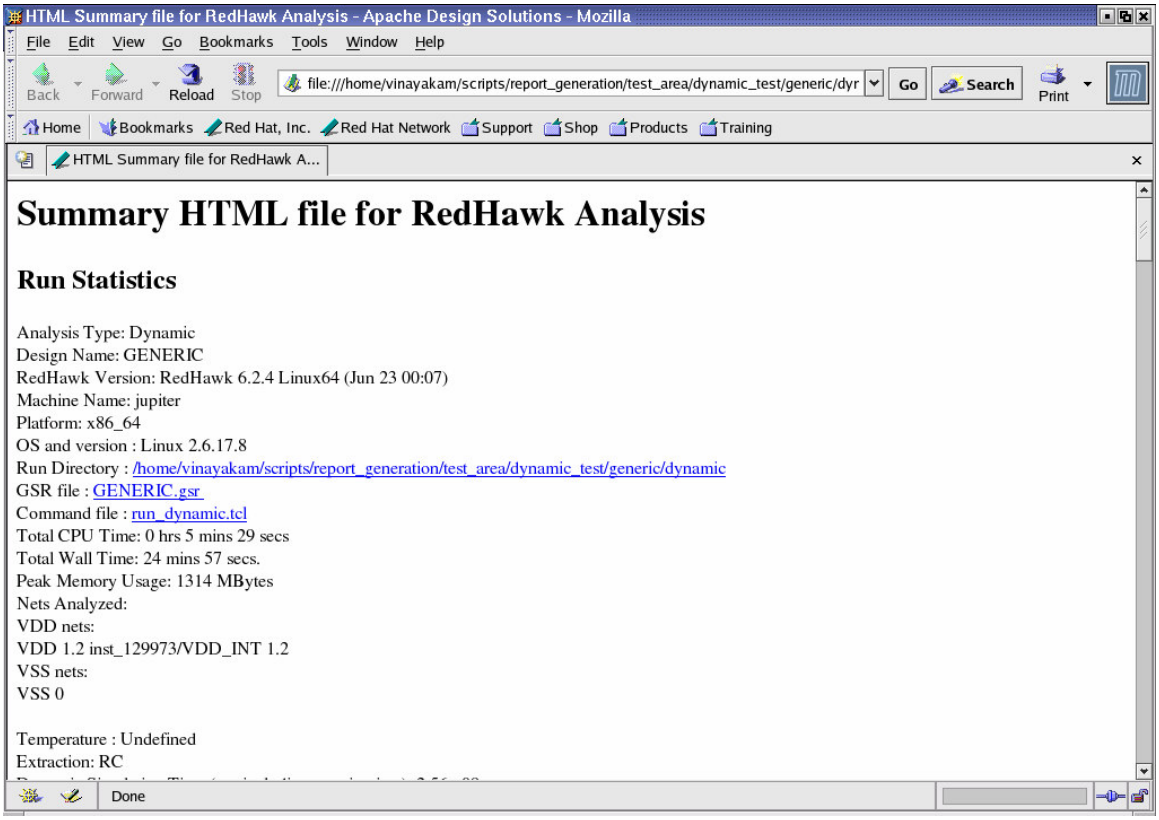
- Resistance Maps
- Decap Maps
- Voltage drop maps for user-defined thresholds

- Movie in animated gif format – for Dynamic and Rampup

Snapshots of Output

Output HTML File

Run Stats



Data Integrity

HTML Summary file for RedHawk Analysis - Apache Design Solutions - Mozilla

Data Integrity Checks

1) Missing LEF definitions: 0

2) Missing LIB definitions: 21
REPORTS:
(a) List of cells present in design, but not in library files:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.refCell.noLib
(b) List of cells with no pins defined in their library definition:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.refCell.noLibPins
(c) List of cells present in design without any internal power tables in their library:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.refCell.noPwr
(d) Lists cells with missing P/G arc information, along with the names of the P/G pins:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.refCell.noPGarcs

3) Missing LEF & LIB definitions: 0

4) SPEF/DSPF Integrity: 1150 mis-matches between SPEF and DEF.
REPORTS:
(a) List of cells with no drivers: /home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.rcNoDriver
(b) Reports of inconsistent drivers between SPEF and DEF files:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.rcMismatch
(c) List of nets and lines in SPEF file that cannot be mapped to a design:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.rcBogus
(d) List of nets with zero capacitance value: /home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.rc0Net

Package,APL, Layout, Errors/Warnings

HTML Summary file for RedHawk Analysis - Apache Design Solutions - Mozilla

Package Vdd Inductance = 2.5 pH
Package Vdd Cap = 5 pF
Package Gnd Resistance = 0.005 ohm
Package Gnd Inductance = 2.5 pH
Package Gnd Cap = 5 pF

8) Apache Power Library:
Number of cells with missing APL current : 29
Number of cells with missing APL cap : 28
REPORTS:
(a) List of cells with no APL current library:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.refCell.noAplCurrent
(b) List of cells with no APL capacitance library:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/apache.refCell.noAplCap
Switch Cell Model Imported: Yes

Design Layout : GENERIC_Layout.gif

Errors/Warnings

Total number of errors : 1
Total number of warnings : 783
REPORTS:
(a) Error report: adsRpt/apache.err.<date_and_timestamp>
(b) Warning report: adsRpt/apache.warn.<date_and_timestamp>

Power Summary

HTML Summary file for RedHawk Analysis - Apache Design Solutions - Mozilla

Power Analysis Summary

Total Chip Power: 0.749997325 W
Clock Network Power: 0.18806 W
Clock Network Power Including ClockPin Power: 0.32006 W
Leakage Power: 0.0003704103 W
Switching Power: 0.459091547 W
Internal Power: 0.2905416 W

REPORTS:
(a) Power Consumption Summary: /home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/power_summary.rpt
(b) Instance-based Power Consumption Report:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/GENERIC.power.rpt
(c) Top 10 instances with highest power consumption:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/reports/dynamic_reports/top10_inst_with_highest_power.rpt

GIF:
(a) Power Density Map: GENERIC_PowerDensity.gif
(b) Instance Power Map: GENERIC_InstPower.gif
(c) Leakage Power Map: GENERIC_LeakagePower.gif
(d) Clock Power Map: GENERIC_ClockPower.gif

Extraction Summary

HTML Summary file for RedHawk Analysis - Apache Design Solutions - Mozilla

Extraction Summary

Instances disconnected from power network: 0
Instances disconnected from ground network: 0
Instanced disconnected from power-ground network: 0
Number of disconnected wires: 9268
Number of disconnected vias: 12630
Number of missing vias: 8855
Number of shorts: 0

REPORTS:
(a) List of instances that are not hooked to the power network:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/GENERIC.power.unused
(b) List of instances belonging to analyzed PG network that are physically isolated from VDD,GND:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/GENERIC.PG.unconnect
These instances will be ignored during dynamic analysis
(c) List of instances belonging to analyzed VDD network that are physically isolated from VDD:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/GENERIC.VDD.unconnect
These instances will be ignored during dynamic analysis
(d) List of instances belonging to analyzed GND network that are physically isolated from GND:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/GENERIC.GND.unconnect
These instances will be ignored during dynamic analysis
(e) List of vias belonging to the net that are physically isolated from a power source: adsRpt/<design_net>.Via.unconnect
(f) List of wires belonging to the net that are physically isolated from the driving source: adsRpt/<design_net>.Wire.unconnect
(g) List of pins belonging to the net that are physically isolated from a power source: adsRpt/<design_net>.Pin.unconnect

Dynamic Analysis summary

HTML Summary file for RedHawk Analysis - Apache Design Solutions - Mozilla

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HTML Summary file for RedHawk A...

Dynamic Simulation Summary

Worst Dynamic Voltage Drop

Worst Wire Dynamic Voltage Drop on VDD net : 195.30mV (Ideal Voltage of 1.200V) at METAL1 (1689.685 , 3875.730)
Worst Wire Dynamic Voltage Drop on GND net : 154.80mV (Ideal Voltage of 0.000V) at METAL1 (1718.465 , 3805.620)
Worst avgTW drop of instances: 346.20mV for instance inst_129228/inst_456444
Worst maxTW drop of instances: 344.50mV for instance inst_129228/inst_455694
Worst minTW drop of instances: 348.40mV for instance inst_129228/inst_461842
Worst minWC drop of instances: 348.60mV for instance inst_129228/inst_462104
Worst peak VDD drop of all instances: 195.3 mV for instance inst_129228/inst_462072
Worst peak VSS bounce of all instances: 154.8 mV for instance inst_129228/inst_461957

Worst peak voltage drop across switch cell : 9.916186e-03 V for switch instance inst_129973/switch_inst_R3_C48
Worst peak internal voltage of switch cell : 1.126621e+00 V for switch instance inst_129973/switch_inst_R1_C52
Maximum peak current through switch cell : 9.919651e-05 A for switch instance inst_129973/switch_inst_R9_C9

REPORTS:
(a) List of all instances worst dynamic voltage drop:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/Dynamic/GENERIC.dvd
(b) List of all instances worst dynamic voltage drop on per-pin basis:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/Dynamic/GENERIC.dvd.pin
(c) List of all instances worst dynamic voltage drop on per-arc basis:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/Dynamic/GENERIC.dvd.arc
(d) Top 10 cells with maximum average dynamic voltage drop:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/reports/dynamic_reports/top10_avgTW_instances.rpt

Dynamic Analysis summary (2)

HTML Summary file for RedHawk Analysis - Apache Design Solutions - Mozilla

(d) Top 10 cells with maximum average dynamic voltage drop:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/reports/dynamic_reports/top10_avgTW_instances.rpt
(e) List of worst EM violation locations:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/Dynamic/GENERIC.em.worst
(f) List of top 5 pad names with highest peak current:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/reports/dynamic_reports/top_5_pads_with_highest_current.rpt
(g) List of all pad names and the current sourced through each pad :
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/Dynamic/pad.current
(h) List of all switch cells and the current, voltage drop across each switch cell:
/home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/adsRpt/Dynamic/switch_dynamic.rpt
(i) Decap Statistics: /home/vinayakam/scripts/report_generation/test_area/dynamic_test/generic/dynamic/reports/dynamic_reports/decap_statistics.rpt

GIF:
(a) Instance dynamic voltage drop maps :
Average over Timing Window : GENERIC_avgTW.gif
Maximum over Timing Window : GENERIC_maxTW.gif
Minimum over Timing Window : GENERIC_minTW.gif
Minimum over Simulation Cycle : GENERIC_minCyc.gif
(b) Worst wire drop map: GENERIC_wireDvD.gif
(c) Peak EM violation map: GENERIC_PeakEM.gif
(d) Peak Pad Current map:GENERIC_PeakPadCurrent.gif
(e) Current map: GENERIC_Current.gif
(f) Decap Density maps: GENERIC_DecapDensity.gif, GENERIC_DeviceDecap.gif, GENERIC_IntentionalDecapDensity.gif, GENERIC_PowerGridParasiticDecap.gif, GENERIC_LoadCap.gif
(g) Dynamic Movie: GENERIC_movie.gif

Output Directory Structure and Contents

```
[vinayakam@jupiter dynamic]$ ls -R reports/
reports/:
dynamic_reports

reports/dynamic_reports:
analysis_summary.html   maps                        statistics.rpt
analysis_summary.rpt    mem_usage_for_each_step.rpt top10_avgTW_instances.rpt
decap_statistics.rpt    movie                       top10_inst_with_highest_power.rpt
gridcheck.rpt           peak_currents_of_pads.rpt   top_5_pads_with_highest_current.rpt

reports/dynamic_reports/maps:
GENERIC_avgTW.gif                   GENERIC_maxTW.gif
GENERIC_ClockPower.gif              GENERIC_minCyc.gif
GENERIC_Current.gif                 GENERIC_minTW.gif
GENERIC_DecapDensity.gif            GENERIC_PeakEM.gif
GENERIC_DeviceDecap.gif             GENERIC_PeakPadCurrent.gif
GENERIC_InstanceToggleRate.gif      GENERIC_PowerDensity.gif
GENERIC_InstFreq.gif                GENERIC_PowerGridParasiticDecap.gif
GENERIC_InstPower.gif               GENERIC_ToggleRateDensity.gif
GENERIC_IntentionalDecapDensity.gif GENERIC_TotalRes.gif
GENERIC_Layout.gif                  GENERIC_VddRes.gif
GENERIC_LeakagePower.gif            GENERIC_VssRes.gif
GENERIC_LoadCap.gif                 GENERIC_wireDvD.gif

reports/dynamic_reports/movie:
GENERIC_movie.gif
```

Rampup Analysis summary

HTML Summary file for RedHawk Analysis - Apache Design Solutions - Mozilla

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Rampup Simulation Summary

Time To Rampup : 5550.000000 picoseconds
Switch instance that turns on last : switch_inst_R17_C52

Peak Current drawn from Battery during Rampup : 6.54721e-06 Amps at time 1300 picoseconds

Off-state voltage of VDD_INT net : 0.098646 V
Off-state leakage of VDD_INT net : 2.944074e-06 A

Worst Dynamic Voltage Drop
Worst Wire Dynamic Voltage Drop on VDD net : 189.30mV (Ideal Voltage of 1.200V) at METAL4 (17.790 , 321.642)
Worst Wire Dynamic Voltage Drop on GND net : 20.30mV (Ideal Voltage of 0.000V) at METAL1 (371.065 , 330.870)

Switch cell statistics:
Name : switch
Type : header
External/Internal Pins : VDD_EXT/VDD_INT
External/Internal Nets : VDD/VDD_INT

Number of switches in the design : 884

Maximum peak current through switch cell : 3.765604e-03 A for switch instance switch_inst_R1_C1

REPORTS:
(a) List of all instances worst dynamic voltage drop:
/home/vinayakam/scripts/report_generation/test_area/rampup_test/generic/send2apache/run/adsRpt/Dynamic/cell_31.dvd

Done

Rampup Analysis summary(2)

HTML Summary file for RedHawk Analysis - Apache Design Solutions - Mozilla

/home/vinayakam/scripts/report_generation/test_area/rampup_test/generic/send2apache/run/adsRpt/Dynamic/switch_dynamic.rpt
(k) Internal node voltage waveform for switch cell with maximum peak current :
/home/vinayakam/scripts/report_generation/test_area/rampup_test/generic/send2apache/run/reports/rampup_reports/switch_inst_R1_C1.header.int
(l) External node voltage waveform for switch cell with maximum peak current :
/home/vinayakam/scripts/report_generation/test_area/rampup_test/generic/send2apache/run/reports/rampup_reports/switch_inst_R1_C1.header.ext

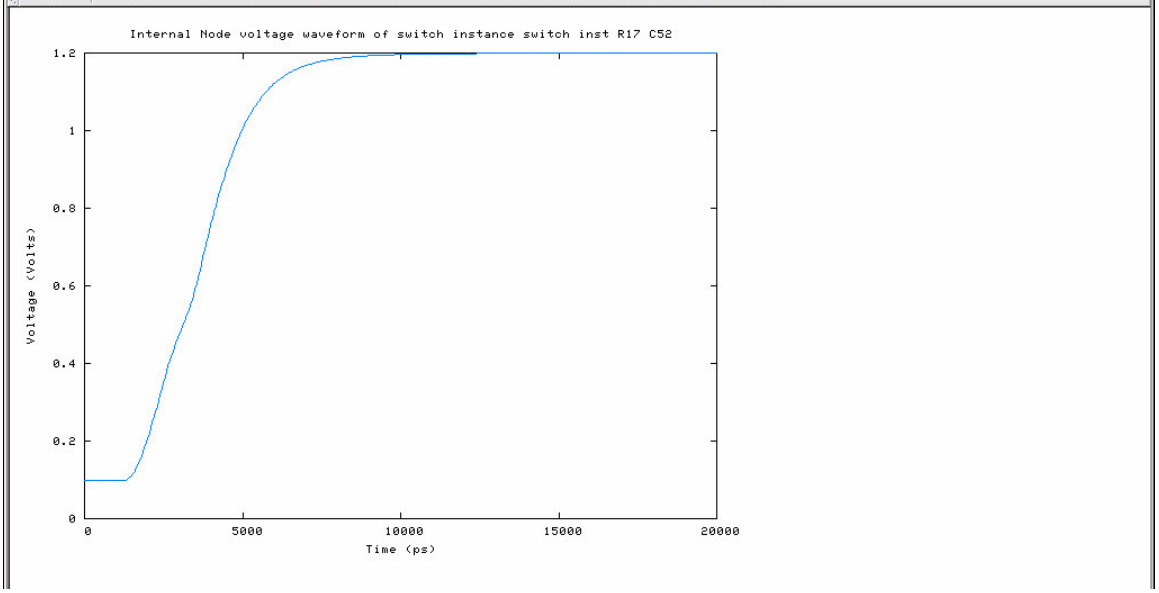
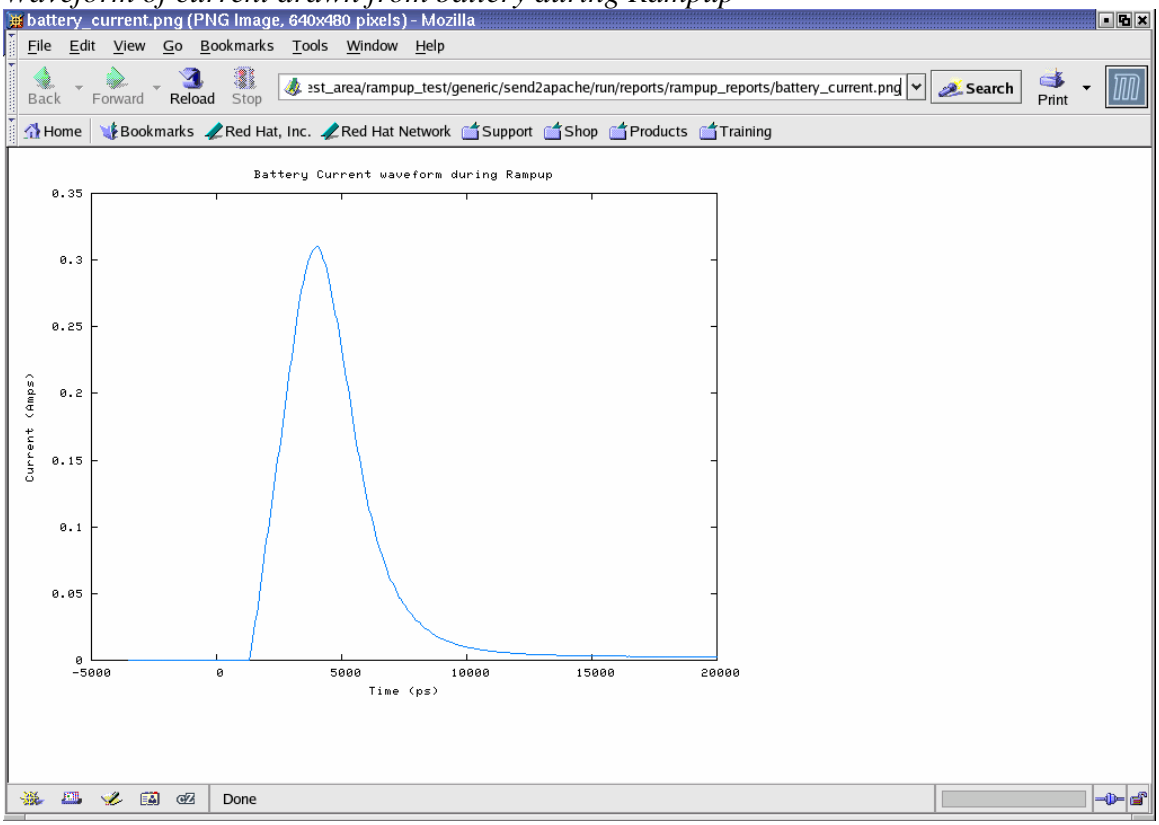
MAPS:
(a) Instance dynamic voltage drop maps :
Average over Timing Window : cell_31_avgTW.gif
Maximum over Timing Window : cell_31_maxTW.gif
Minimum over Timing Window : cell_31_minTW.gif
Minimum over Simulation Cycle : cell_31_minCyc.gif
(b) Worst wire drop map: cell_31_wireDvD.gif
(c) Peak EM violation map: cell_31_PeakEM.gif
(d) Peak Pad Current map:cell_31_PeakPadCurrent.gif
(e) Current map: cell_31_Current.gif
(f) Decap Density maps: cell_31_DecapDensity.gif, cell_31_DeviceDecap.gif, cell_31_IntentionalDecapDensity.gif, cell_31_PowerGridParasiticDecap.gif, cell_31_LoadCap.gif

MOVIE:
(a) Rampup Movie: cell_31_movie.gif

WAVEFORMS:
(a) Internal node voltage waveform for switch instance that turns on last :
/home/vinayakam/scripts/report_generation/test_area/rampup_test/generic/send2apache/run/reports/rampup_reports/last_switch.png
(b) Waveform of Total current drawn from the battery :
/home/vinayakam/scripts/report_generation/test_area/rampup_test/generic/send2apache/run/reports/rampup_reports/battery_current.png

Internal node voltage rampup waveform of last switch

last_switch.png (PNG Image, 640x480 pixels) - Mozilla



Waveform of current drawn from battery during Rampup